



# Automate Checkmk with Ansible

# About me

- René Koch
- Self-employed consultant for:
  - Red Hat Ansible (Automation Platform)
  - Red Hat Enterprise Linux
  - Red Hat Satellite
  - Red Hat Identity Management (IPA)
- Experienced monitoring user (Nagios, Icinga, Checkmk)



# About me

- René Koch
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**Please introduce yourself**

# Time table

- 09:00 – 10:30: Workshop
- 10:30 – 10:45: Break
- 10:45 – 12:30: Workshop
- 12:30 – 13:45: Lunch break
- 13:45 – 15:30: Workshop
- 15:30 – 15:45: Break
- 15:45 – 17:15: Workshop

# Table of content

- Prepare LAB environment
- Installation of Checkmk server
- Installation of Checkmk agents
- Checkmk server configuration
- Fetching information with lookup plugins
- Using Checkmk as Ansible Inventory
- Demo: Self healing with Checkmk, Ansible Automation Platform and Rule books



# Preparing the LAB environment

# Preparing the LAB environment

- Create LAB environment
- Ansible control node =  
Checkmk server =  
Checkmk agent
- Use local connection
- Configure sudo for privileged access



# Preparing the LAB environment

- Slides and code are available on Github:  
[https://github.com/rk-it-at/checkmk\\_conference\\_2026](https://github.com/rk-it-at/checkmk_conference_2026)
- Optional: Install and clone Repository:

```
git clone https://github.com/rk-it-at/checkmk_conference_2026.git
```

# Preparing the LAB environment

- Available VMs (if you don't have your own)
- Passwords: <TODO>

# Preparing the LAB environment

| Hostname     | IP Address | Username | Participant |
|--------------|------------|----------|-------------|
| ansible-ws-1 |            | ansible  |             |
| ansible-ws-2 |            | ansible  |             |
| ansible-ws-3 |            | ansible  |             |
| ansible-ws-4 |            | ansible  |             |
| ansible-ws-5 |            | ansible  |             |
| ansible-ws-6 |            | ansible  |             |
| ansible-ws-7 |            | ansible  |             |

# Preparing the LAB environment

| Hostname      | IP Address | Username | Participant |
|---------------|------------|----------|-------------|
| ansible-ws-8  |            | ansible  |             |
| ansible-ws-9  |            | ansible  |             |
| ansible-ws-10 |            | ansible  |             |
| ansible-ws-11 |            | ansible  |             |
| ansible-ws-12 |            | ansible  |             |
| ansible-ws-13 |            | ansible  |             |
| ansible-ws-14 |            | ansible  |             |
| ansible-ws-15 |            | ansible  |             |

# Preparing the LAB environment

- Configure sudo for privileged access

```
$ su -  
# <EDITOR> /etc/sudoers.d/ansible
```

```
<USERNAME> ALL=(ALL) NOPASSWD: ALL
```

## NOTE

Use user root to run this command!

Replace <EDITOR> with vi, nano or your favorite editor

Replace <USERNAME> with your system user

# LAB 1: Configure sudo

# LAB 1: Configure sudo

- Configure sudo for privileged access

```
$ su -  
# <EDITOR> /etc/sudoers.d/ansible
```

```
<USERNAME> ALL=(ALL) NOPASSWD: ALL
```

## NOTE

Use user root to run this command!

Replace <EDITOR> with vi, nano or your favorite editor

Replace <USERNAME> with your system user

# Install Ansible





# Install Ansible

- Most Linux distributions ship 2 versions:
  - **ansible-core**: Ansible binary + minimal set of collections
  - **ansible**: Ansible and selected collections

# Install Ansible - RHEL

- Install Ansible on RHEL 10

```
subscription-manager repos --enable codeready-builder-for-rhel-10-$(arch)-rpms  
dnf install https://dl.fedoraproject.org/pub/epel/epel-release-latest-10.noarch.rpm  
dnf install ansible
```

# Install Ansible – RHEL derivatives

- Install Ansible on AlmaLinux, Rocky Linux 10

```
dnf config-manager --set-enabled crb  
dnf install epel-release  
dnf install ansible
```

# Install Ansible – openSUSE

- Install Ansible on openSUSE

```
zypper install ansible
```

# Install Ansible – Ubuntu

- Install Ansible on Ubuntu 24.04+

```
apt update  
apt install ansible
```

# Install Ansible – pip

- Install Ansible with pip

```
pip install ansible
```

# LAB 2: Install Ansible

## LAB 2: Install Ansible

- Ensure Ansible is installed

```
ansible --version
```

```
ansible [core 2.16.18]
```

### NOTE

Depending on your operating system the Ansible version can be different





# Prepare Ansible inventory

# Prepare Ansible inventory

- Create folder for your Ansible code

```
mkdir playbooks
cd playbooks
<EDITOR> hosts
cmk-server ansible_connection=local
cmk-agent ansible_connection=local
```

## NOTE

Replace <EDITOR> with vi, nano or your favorite editor

# Test connection to target server

- Make sure your target server is reachable

```
ansible all -i hosts -m ping
cmk-server | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
cmk-agent | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```

Ansible does not send an ICMP ping to test the connection, it tries to log into the target machine.



# LAB 3: Add target to Ansible inventory

# LAB 3: Add target to Ansible inventory

- Make sure your target server is reachable

```
$ ansible all -i hosts -m ping
cmk-server | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
cmk-agent | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```



# Install Checkmk Collection

# Install Checkmk Collections

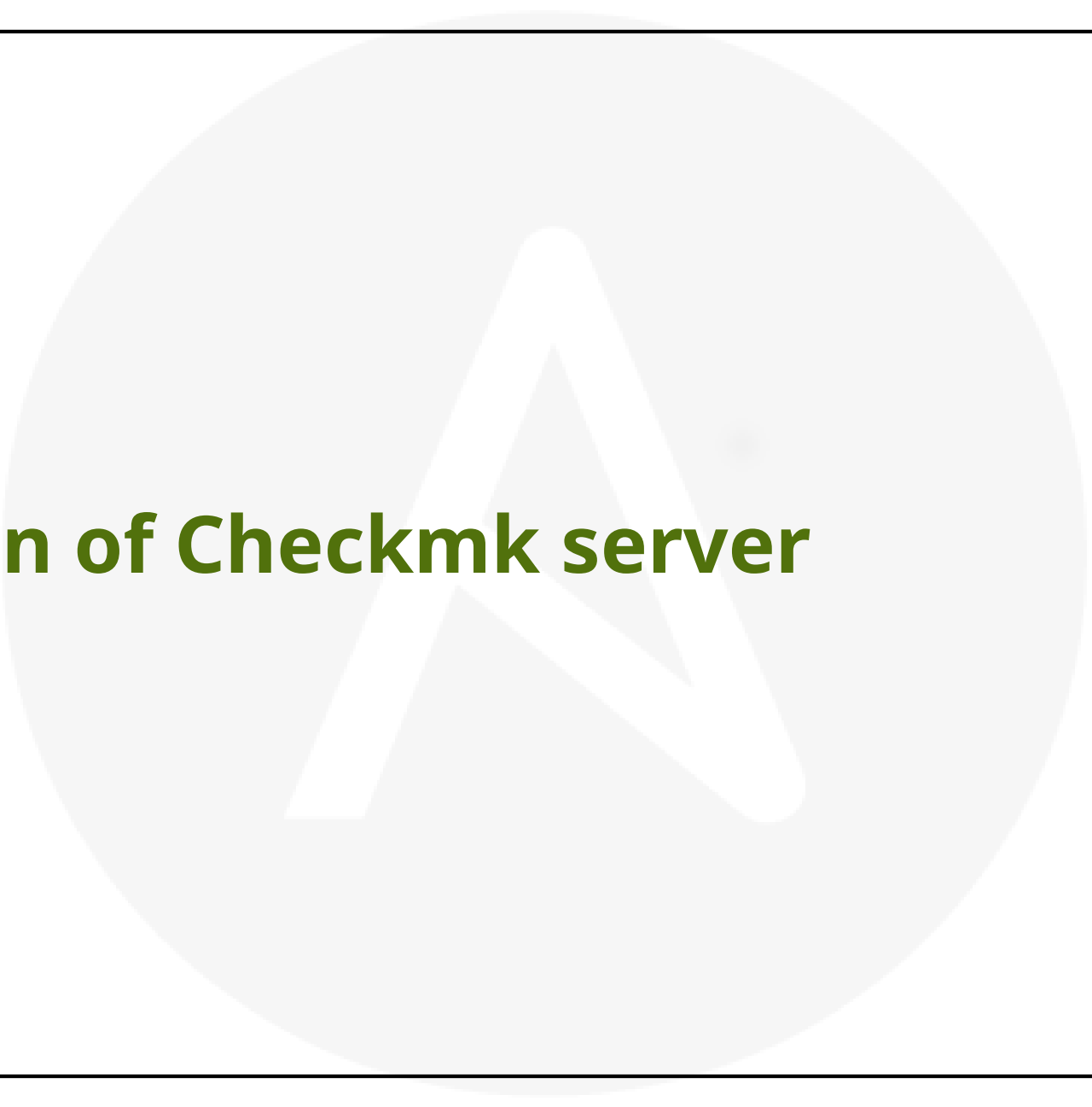
- Simplified and consistent content delivery
- Collections contains:
  - Modules
  - Playbooks
  - Roles
  - Plugins
  - Docs
  - Tests

# Install Checkmk Collection

- Development:  
<https://github.com/Checkmk/ansible-collection-checkmk.general>
- Hosted at Galaxy:  
<https://galaxy.ansible.com/ui/repo/published/checkmk/general>
- Install Collection:

```
ansible-galaxy collection install checkmk.general
```





# Installation of Checkmk server

# Installation of Checkmk server

- Can be installed with provided role server
- Strongly suggest to write own role to fit into own Ansible environment and coding guidelines!
- Documentation for role:  
<https://galaxy.ansible.com/ui/repo/published/checkmk/general/content/role/server/>

# Installation of Checkmk server

- Create playbook

```
<EDITOR> cmk_server.yml
```

# Installation of Checkmk server

```
---
```

```
- name: Install Checkmk server
  hosts: cmk-server
  connection: local
  become: true

  roles:
    - checkmk.general.server
```

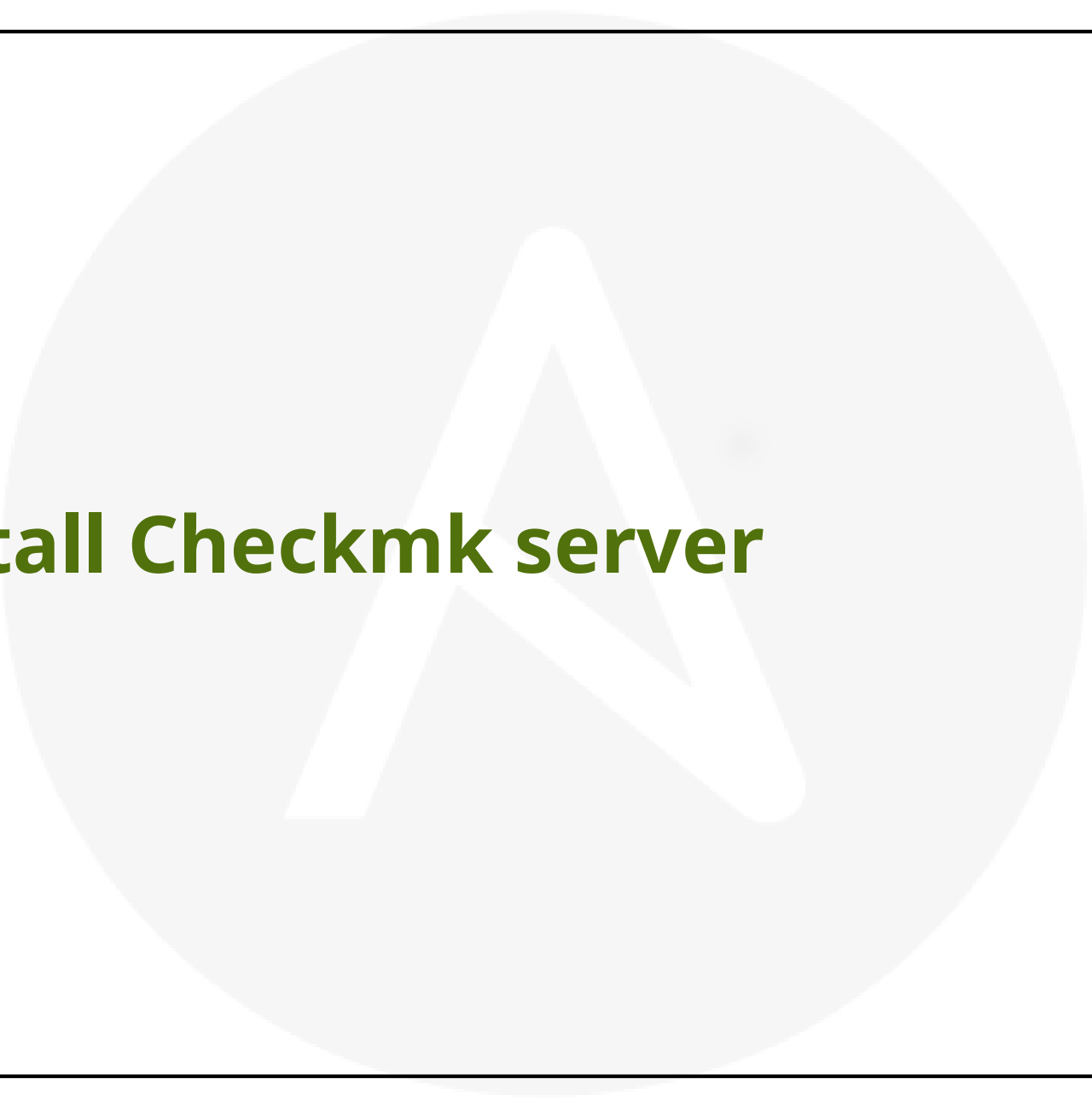
# Installation of Checkmk server

- Create required variables

```
mkdir host_vars  
<EDITOR> host_vars/cmks-server.yml
```

# Installation of Checkmk server

```
---
checkmk_server_version: "2.5.0"
checkmk_server_edition: "community"
checkmk_admin_pw: "AnsibleWOrkshop2026!"
checkmk_server_sites:
  - name: "master"
    version: "{{ checkmk_server_version }}"
    update_conflict_resolution: "abort"
    state: started
    admin_pw: "{{ checkmk_admin_pw }}"
```



# LAB 4: Install Checkmk server

# LAB 4: Install Checkmk server

- Ensure Checkmk server is installed

```
ansible-playbook -i hosts cmk_server.yml

[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17

PLAY [Install Checkmk server] *****

TASK [Gathering Facts] *****
ok: [cmk-server]

TASK [checkmk.general.server : Flush Handlers.] *****

...

PLAY RECAP *****
cmk-server : ok=20 changed=8 unreachable=0 failed=0 skipped=21 rescued=0 ignored=0
```



# Installation of Checkmk server

- Ensure HTTPS is configured

```
<EDITOR> cmk_server.yml
```

# Installation of Checkmk server

```
...
tasks:
  - name: Install mod_ssl for https access
    ansible.builtin.dnf:
      name: mod_ssl
      state: installed
      notify: Restart httpd

handlers:
  - name: Restart httpd
    ansible.builtin.service:
      name: httpd
      state: restarted
```



# LAB 5: Secure Checkmk server

## LAB 5: Secure Checkmk server

- Ensure Checkmk server is listening to HTTPS

```
ansible-playbook -i hosts cmk_server.yml
```

```
PLAY RECAP *****
cmk-server : ok=17 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```

- Access your server: `https://<ip_address>/master`
- Username: `cmkadmin`
- Password: `AnsibleW0rkshop2026!`





# Installation of Checkmk agent

# Installation of Checkmk agent

- Can be installed with provided role agent
- Strongly suggest to write own role to fit into own Ansible environment and coding guidelines!
- Documentation for role:  
<https://galaxy.ansible.com/ui/repo/published/checkmk/general/content/role/agent/>

# Installation of Checkmk agent

- Create playbook

```
<EDITOR> cmk_agent.yml
```



# Installation of Checkmk agent

```
---  
  
- name: Install Checkmk agent  
  hosts: cmk-agent  
  connection: local  
  become: true  
  
  pre_tasks:  
    - name: Install required Python packages  
      ansible.builtin.dnf:  
        name: python3-netaddr  
        state: installed  
  
  roles:  
    - checkmk.general.agent
```

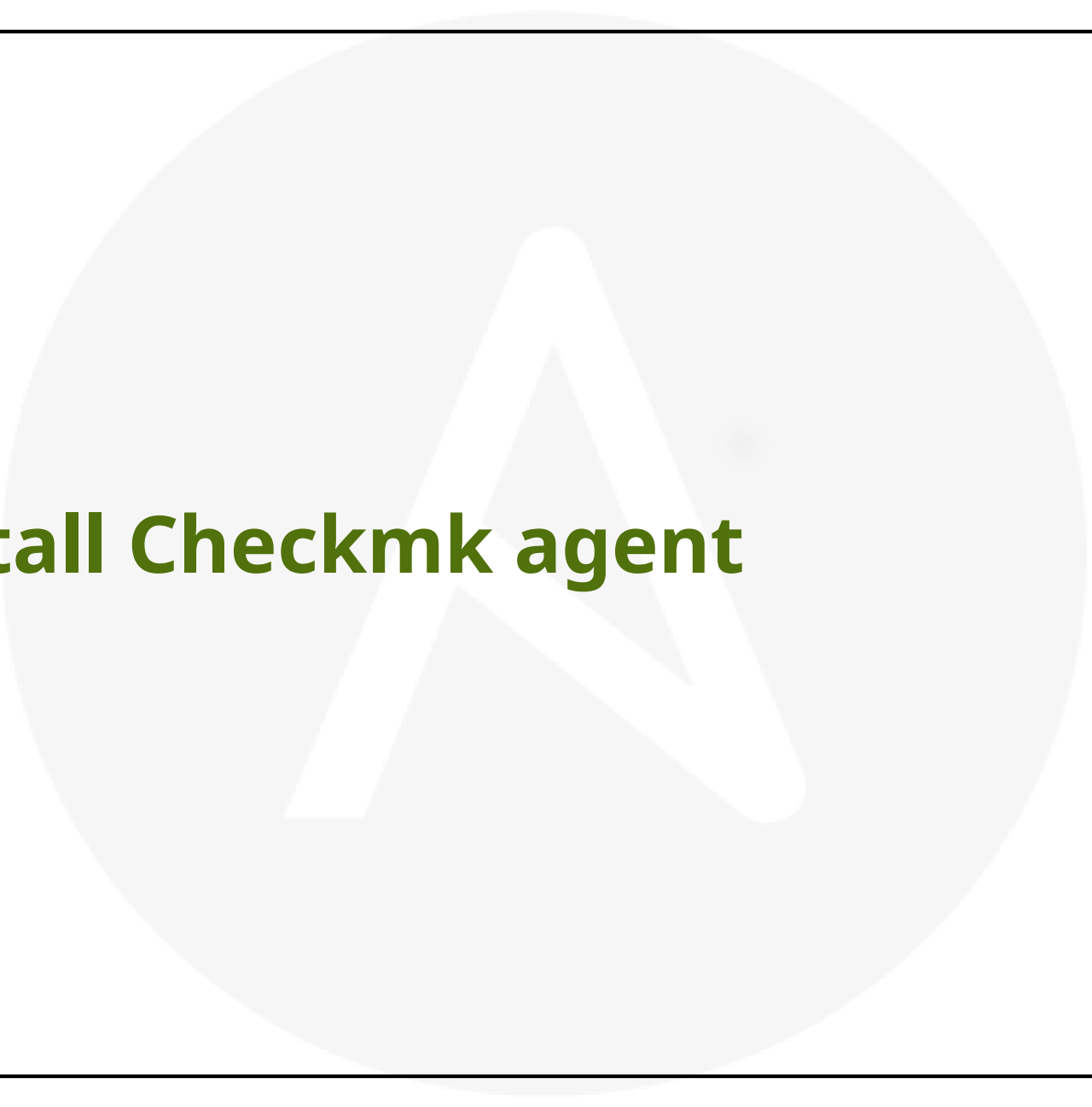
# Installation of Checkmk agent

- Create required variables

```
mkdir host_vars  
<EDITOR> host_vars/cmk-agent.yml
```

# Installation of Checkmk agent

```
---  
checkmk_agent_version: "2.5.0"  
checkmk_agent_edition: "community"  
checkmk_agent_server: localhost  
checkmk_agent_server_validate_certs: "false"  
checkmk_agent_site: "master"  
checkmk_agent_user: "cmkadmin"  
checkmk_agent_pass: "AnsibleW0rkshop2026!"
```



# LAB 6: Install Checkmk agent

# LAB 6: Install Checkmk agent

- Ensure Checkmk agent is installed

```
$ ansible-playbook -i hosts cmk_agent.yml

[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17

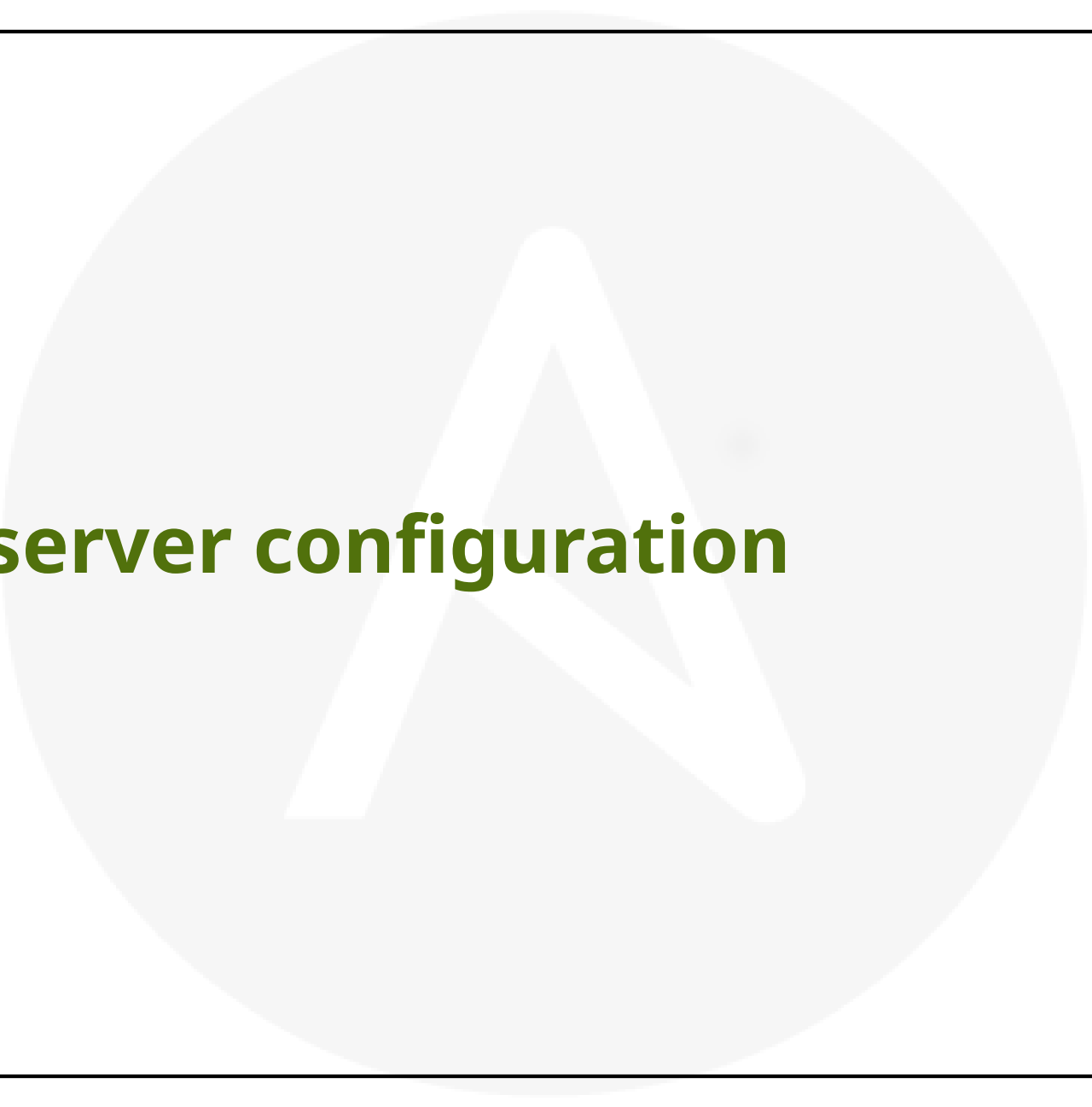
PLAY [Install Checkmk agent] *****

TASK [Gathering Facts] *****
ok: [cmk-agent]

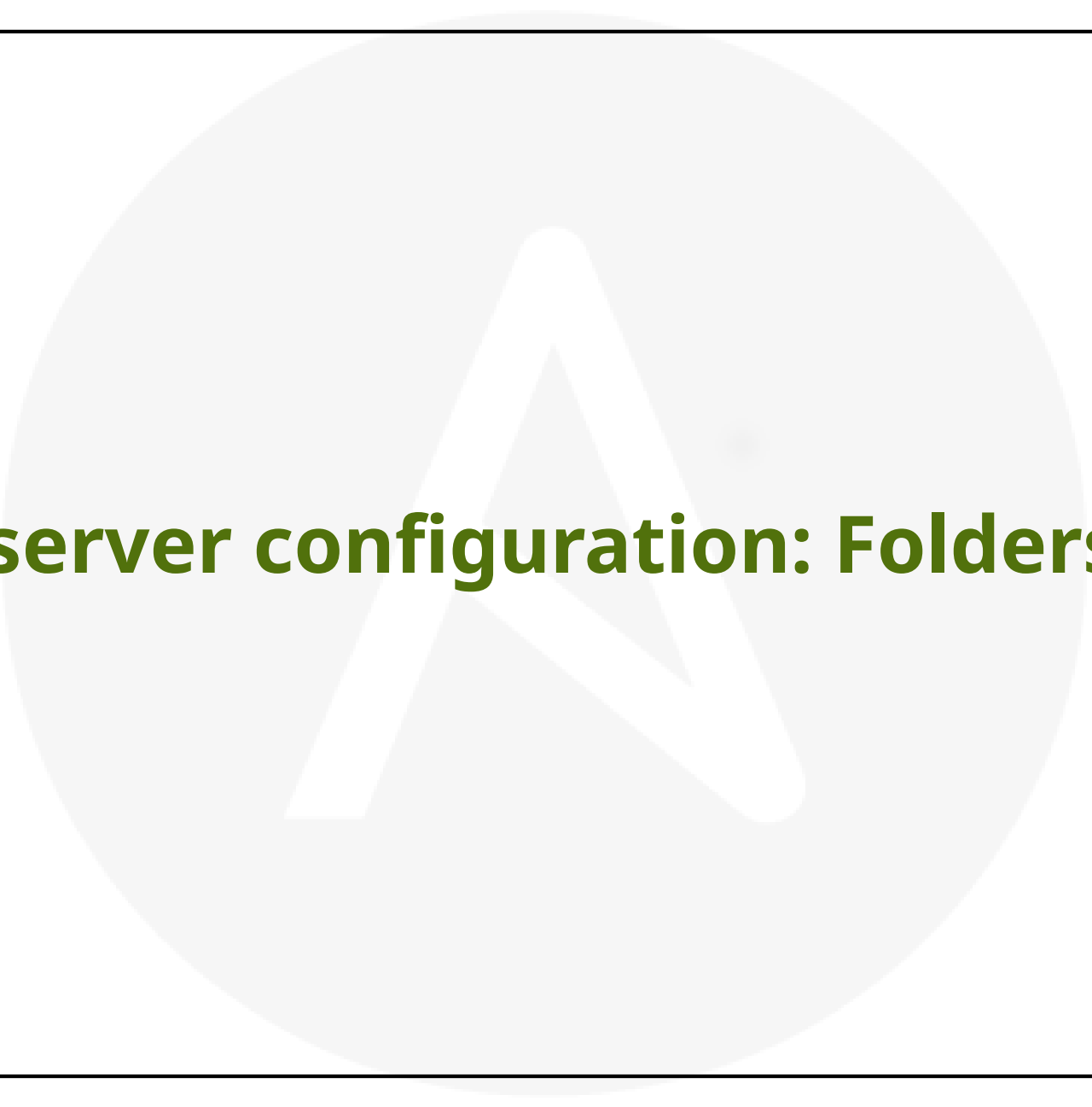
...

TASK [checkmk.general.agent : Update monitored services and labels on host.] *****
ok: [cmk-agent]

PLAY RECAP *****
cmk-agent : ok=19 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



# Checkmk server configuration



# Checkmk server configuration: Folders

# Checkmk configuration: Folders

- Create 2 folders - **/linux** and **/linux/checkmk**
- Add **Label** to folder linux and ensure **Checkmk agent** is used for monitoring

```
<EDITOR> host_vars/cmkn-server.yml
```



# Checkmk configuration: Folders

```
...
checkmk_folders:
- path: /linux
  attributes:
    labels:
      os: almalinux
      tag_agent: "cmk-agent"
- path: /linux/checkmk
```

## NOTE

Create folder in WebUI first, and access API to get syntax for attributes!

# Checkmk configuration: Folders

- Create 2 folders - **/linux** and **/linux/checkmk**
- Add **Label** to folder linux and ensure **Checkmk agent** is used for monitoring
- Append new task to existing ones

```
<EDITOR> cmk_server.yml
```

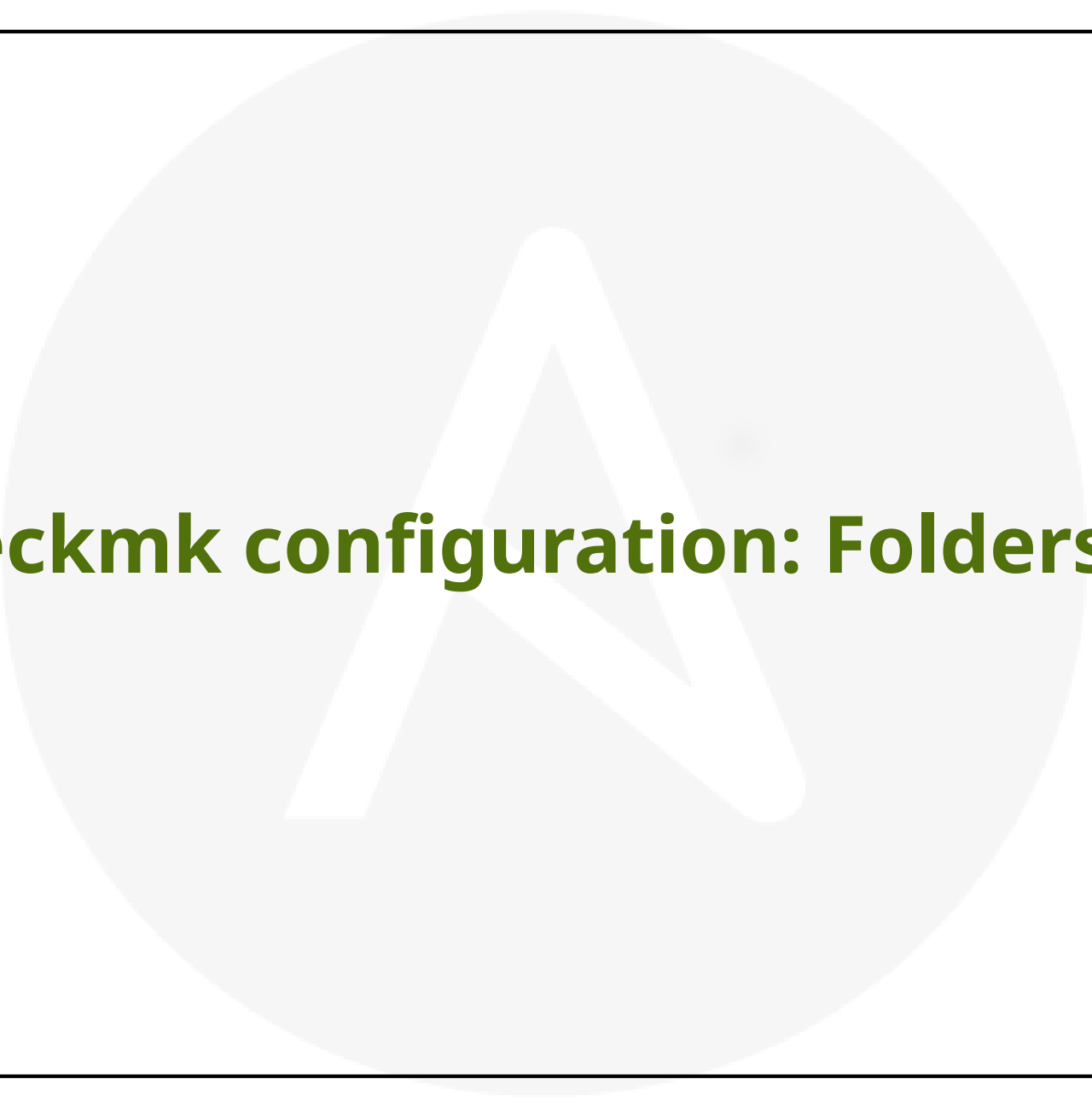
# Checkmk configuration: Folders

```
...  
- name: Create folder  
  checkmk.general.folder:  
    name: "{{ item['name'] | default(omit) }}"  
    path: "{{ item['path'] }}"  
    attributes: "{{ item['attributes'] | default(omit) }}"  
    state: present  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"  
    loop: "{{ checkmk_folders }}"
```

# Checkmk configuration: Folders

- Append new handler

```
...  
- name: Activate Checkmk changes  
  checkmk.general.activation:  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    force_foreign_changes: true  
    run_once: true
```



# LAB 7: Checkmk configuration: Folders

# LAB 7: Checkmk configuration: Folders

- Ensure folder is created

```
ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17
```

```
PLAY [Install Checkmk server] *****
```

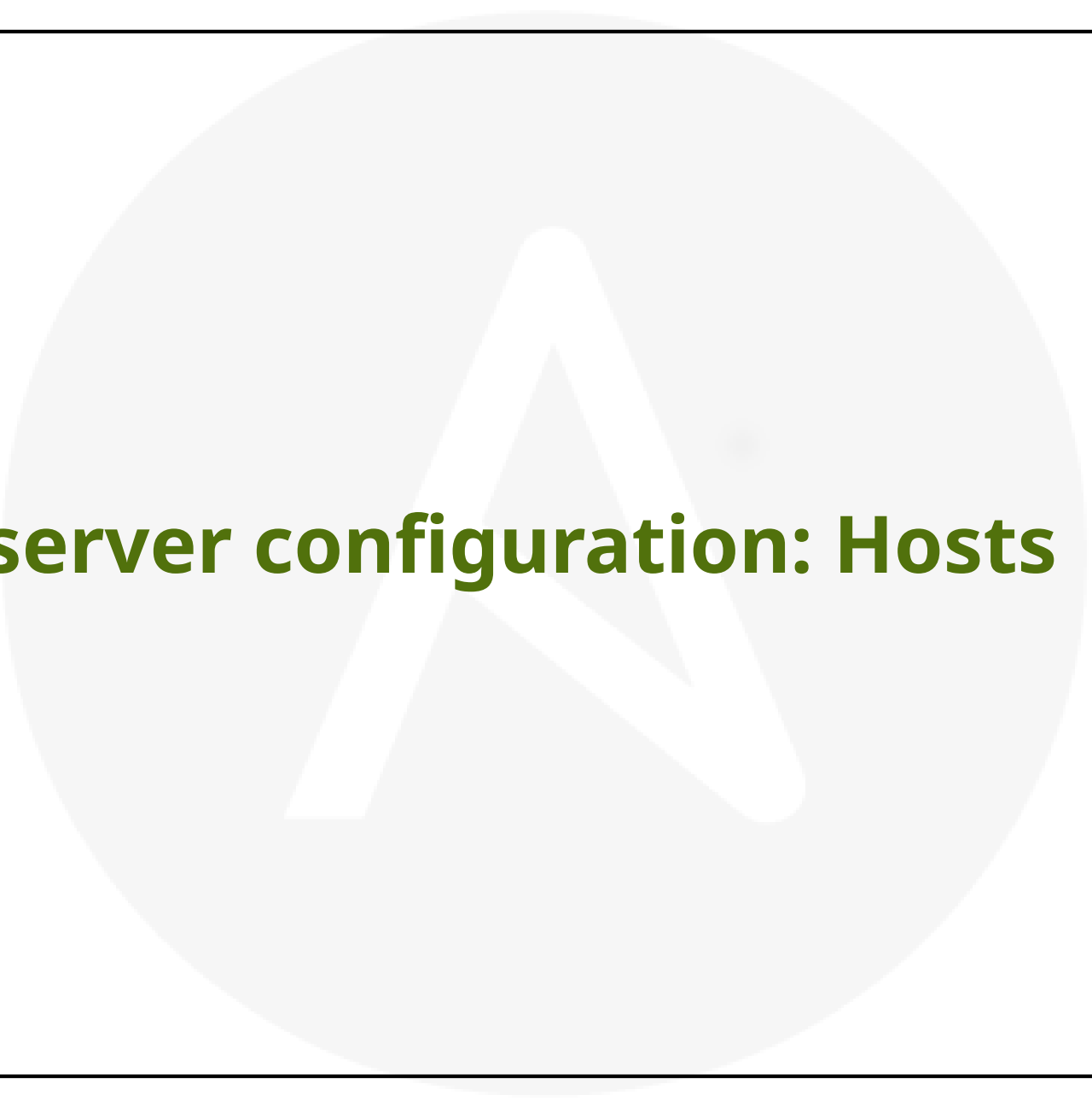
```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=19 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



# Checkmk server configuration: Hosts

# Checkmk configuration: Hosts

- Create host **cmk-agent** in folder **/linux/checkmk** with ip **127.0.0.1**

```
<EDITOR> host_vars/cmk-agent.yml
```



# Checkmk configuration: Hosts

```
...  
checkmk_host_settings:  
  folder: /linux/checkmk  
  attributes:  
    ipaddress: "127.0.0.1"
```

## NOTE

Create host in WebUI first, and access API to get syntax for attributes!

# Checkmk configuration: Hosts

- Create host **cmk-agent** in folder **/linux/checkmk** with ip **127.0.0.1**
- Add new task

```
$ <EDITOR> cmk_agent.yml
```

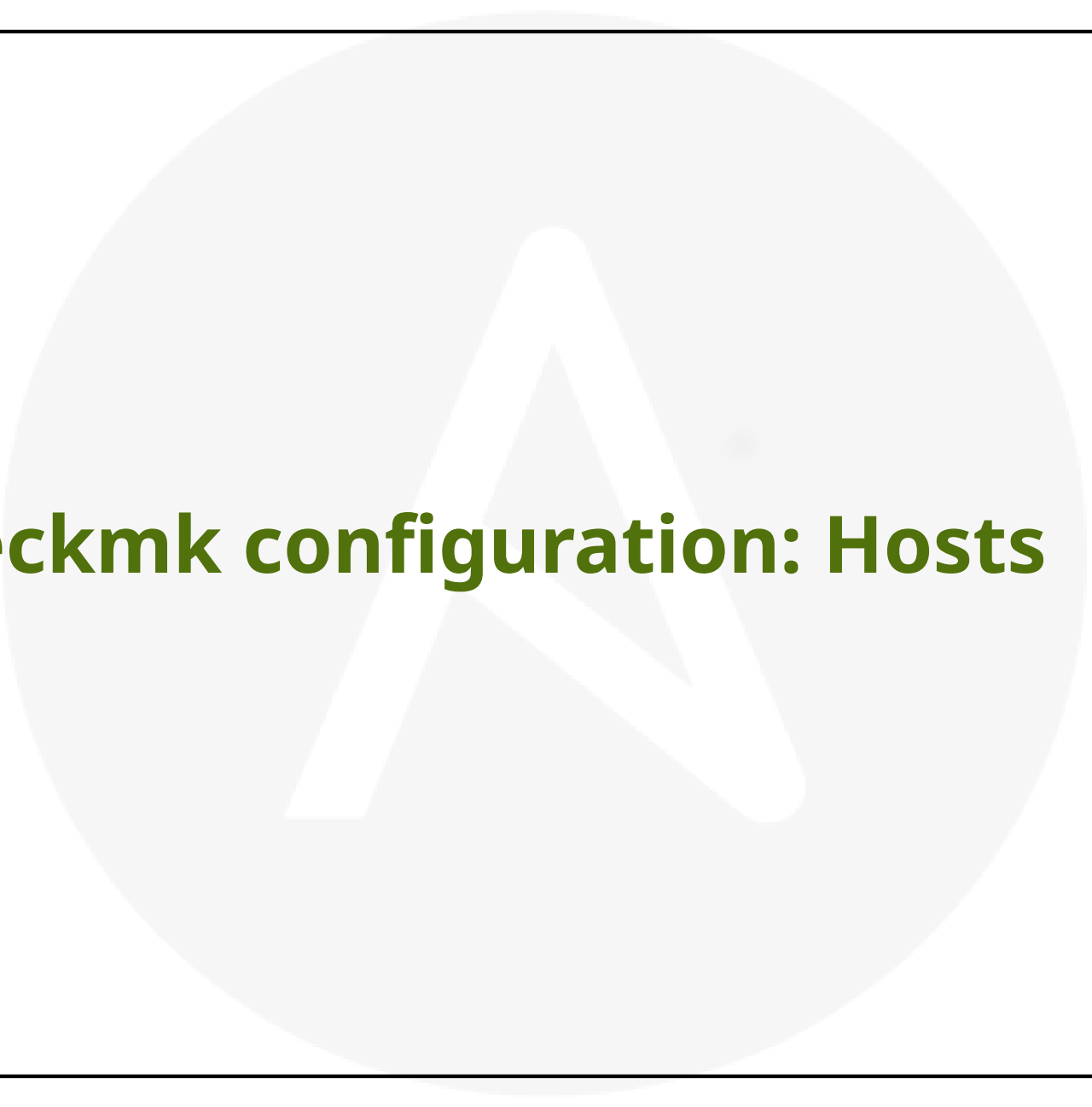
# Checkmk configuration: Hosts

```
...
tasks:
- name: Create host
  checkmk.general.host:
    name: "{{ inventory_hostname }}"
    folder: "{{ checkmk_host_settings['folder'] | default(omit) }}"
    attributes: "{{ checkmk_host_settings['attributes'] | default(omit) }}"
    state: present
    server_url: "https://localhost"
    site: "master"
    automation_user: "cmkadmin"
    automation_secret: "{{ checkmk_agent_pass }}"
    validate_certs: false
  notify: "Activate Checkmk changes"
  loop: "{{ checkmk_host_settings | dict2items }}"
```

# Checkmk configuration: Hosts

- Append new handler

```
...
handlers:
- name: Activate Checkmk changes
  checkmk.general.activation:
    server_url: "https://localhost"
    site: "master"
    automation_user: "cmkadmin"
    automation_secret: "{{ checkmk_agent_pass }}"
    validate_certs: false
    force_foreign_changes: true
    run_once: true
```



# LAB 8: Checkmk configuration: Hosts

# LAB 8: Checkmk configuration: Hosts

- Ensure host is created

```
$ ansible-playbook -i hosts cmk_agent.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17
```

```
PLAY [Install Checkmk agent] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-agent]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-agent : ok=21 changed=2 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



# Checkmk server configuration: Service discovery

# Checkmk configuration: Discovery

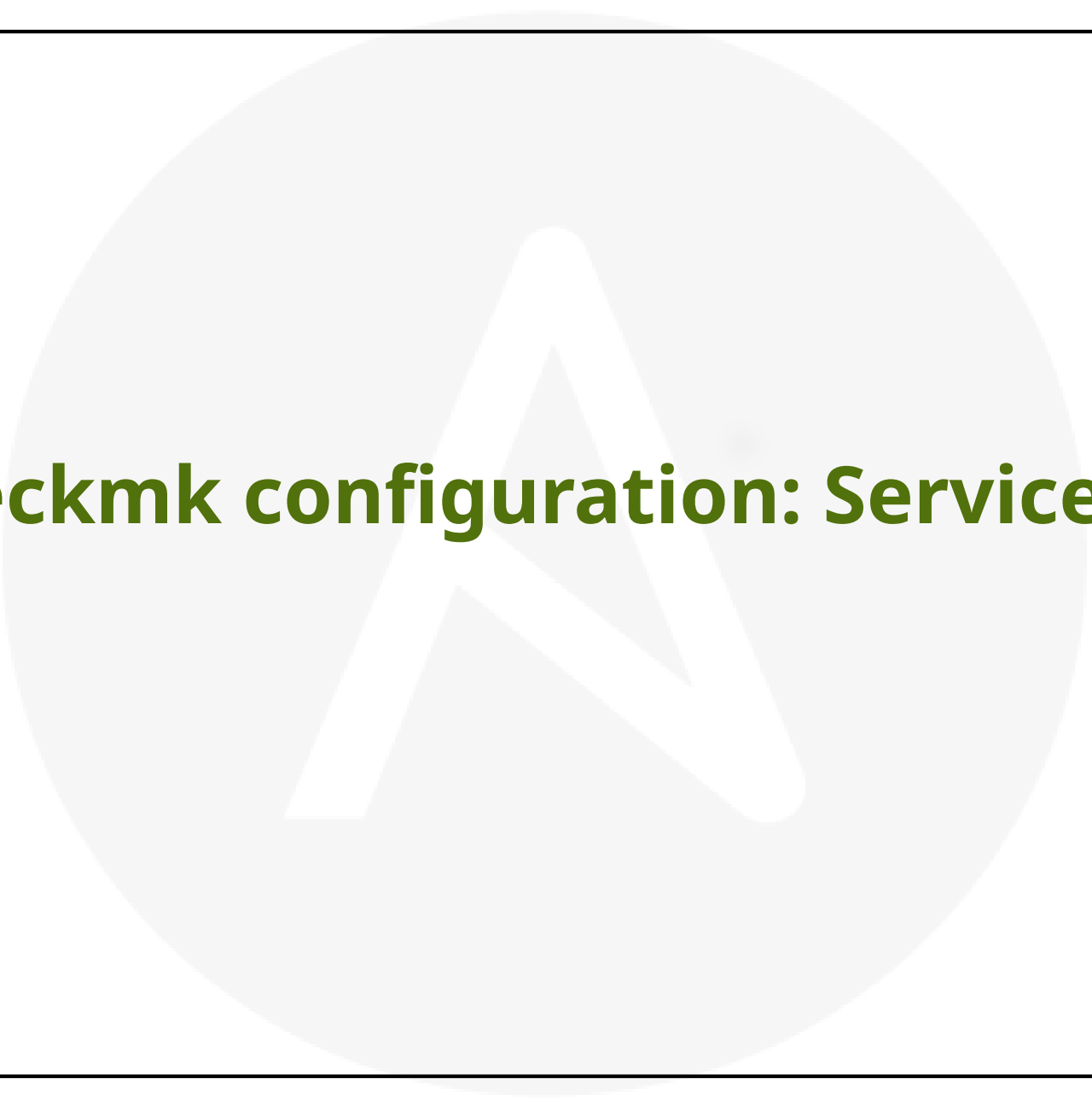
- Run service discovery and add **unmonitored** services
- Append task to existing ones

```
<EDITOR> cmk_agent.yml
```



# Checkmk configuration: Discovery

```
...  
- name: Run service discovery  
  checkmk.general.discovery:  
    host_name: "{{ inventory_hostname }}"  
    state: new  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_agent_pass }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"
```



# LAB 9: Checkmk configuration: Service discovery

# LAB 9: Checkmk configuration: Discovery

- Ensure services are monitored

```
ansible-playbook -i hosts cmk_agent.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17
```

```
PLAY [Install Checkmk agent] *****
```

```
TASK [Gathering Facts] *****
```

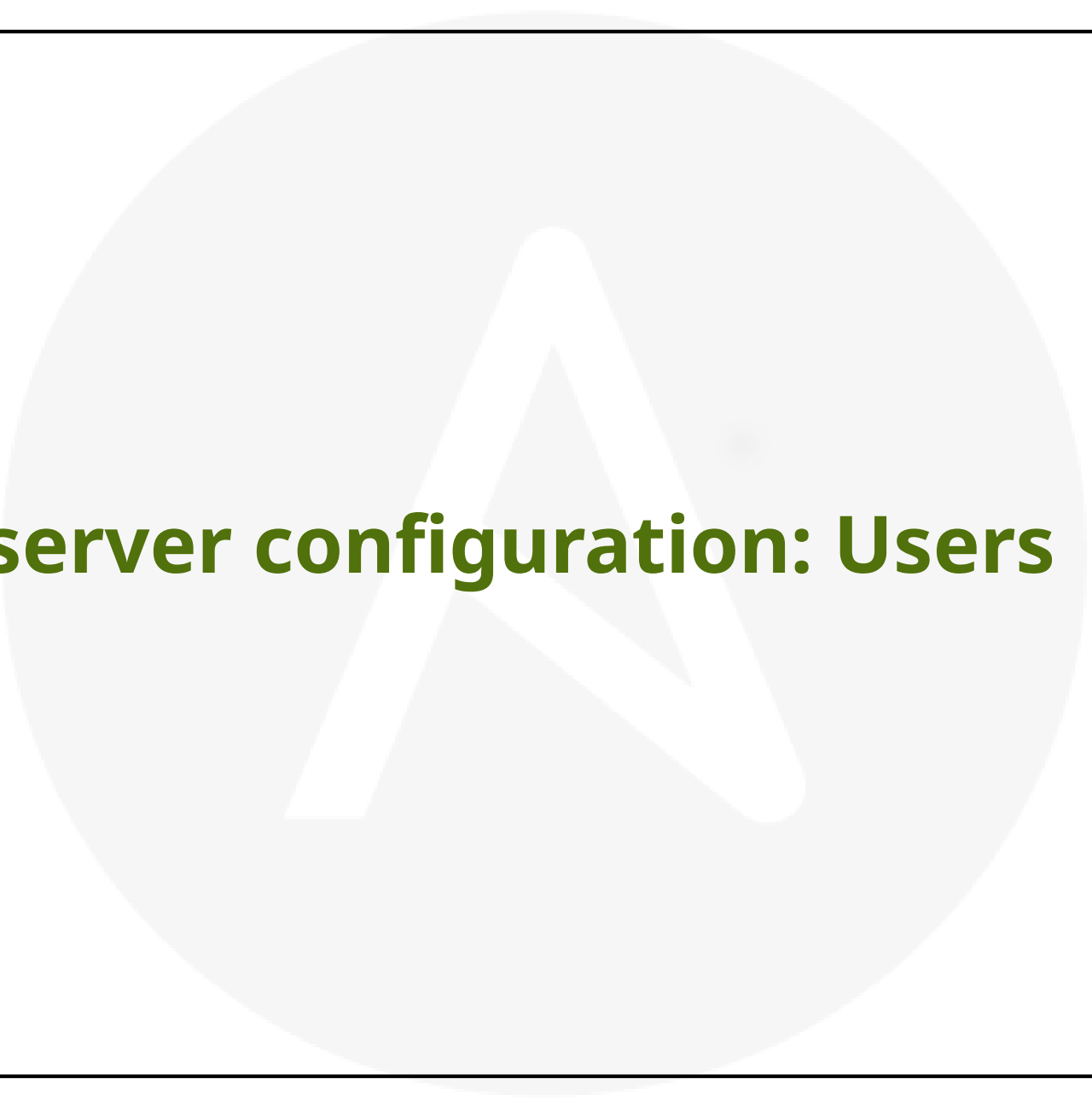
```
ok: [cmk-agent]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-agent : ok=21 changed=2 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```





# Checkmk server configuration: Users

# Checkmk configuration: Users

- Create user **cmkuser** with email address **cmkuser@localhost** and member of contact group **Everything** and role **Normal user**

```
<EDITOR> host_vars/cmk-server.yml
```

# Checkmk configuration: Users

```
...
checkmk_users:
  - name: cmkuser
    password: "AnsibleW0rkshop2026!"
    email: "cmkuser@localhost"
    fullname: "Checkmk user"
    auth_type: password
    contactgroups:
      - "all"
    roles:
      - "user"
```

# Checkmk configuration: Users

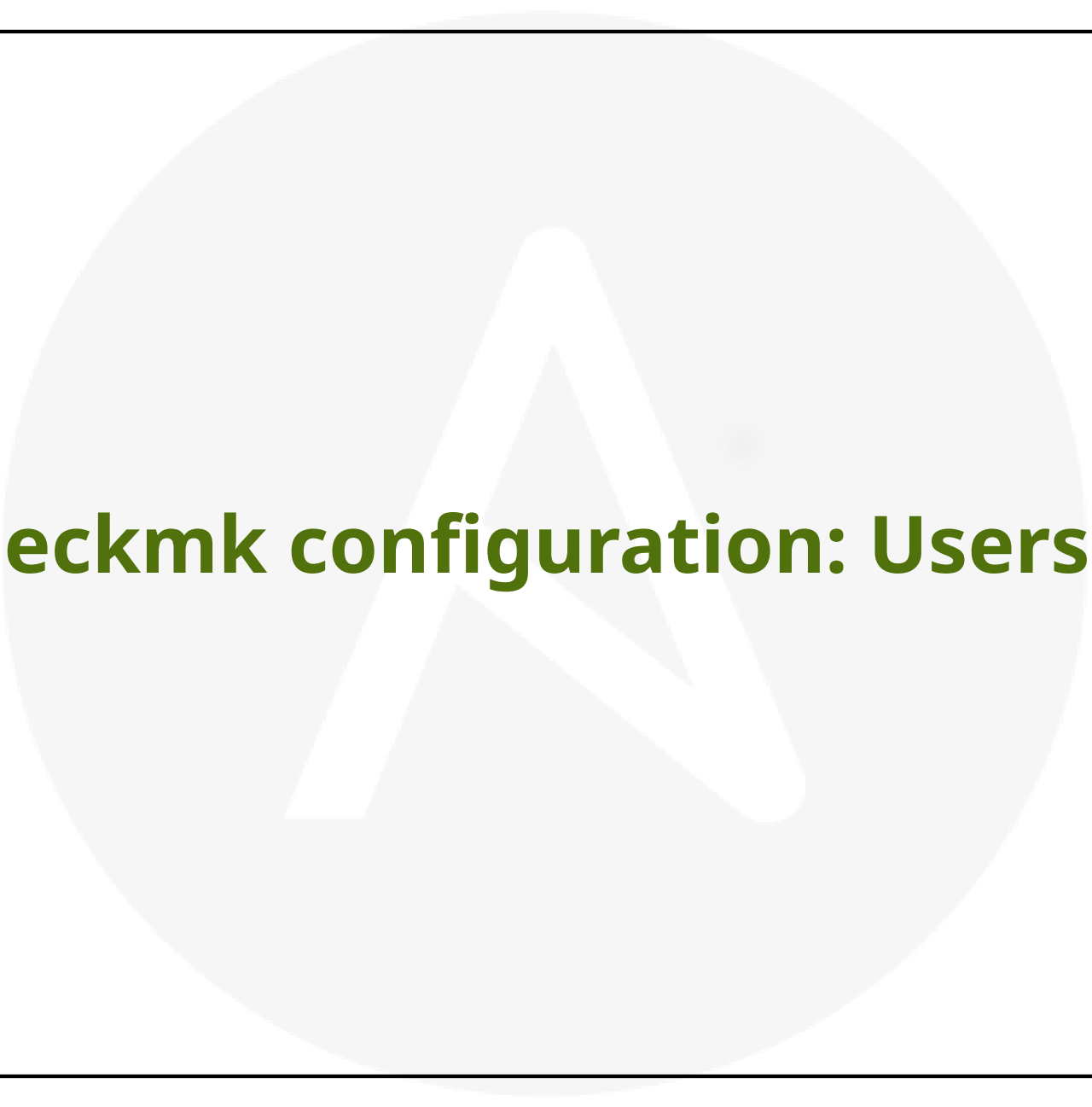
- Create user **cmkuser** with email address **cmkuser@localhost** and member of contact group **Everything** and role **Normal user**
- Add new task after existing ones

```
<EDITOR> cmk_server.yml
```



# Checkmk configuration: Users

```
...
- name: Create user
  checkmk.general.user:
    name: "{{ item['name'] }}"
    password: "{{ item['password'] | default(omit) }}"
    email: "{{ item['email'] | default(omit) }}"
    fullname: "{{ item['fullname'] | default(omit) }}"
    auth_type: "{{ item['auth_type'] }}"
    contactgroups: "{{ item['contactgroups'] | default(omit) }}"
    roles: "{{ item['roles'] | default(omit) }}"
    state: present
    server_url: "https://localhost"
    site: "master"
    automation_user: "cmkadmin"
    automation_secret: "{{ checkmk_admin_pw }}"
    validate_certs: false
    notify: "Activate Checkmk changes"
    loop: "{{ checkmk_users }}"
```



# LAB 10: Checkmk configuration: Users

# LAB 10: Checkmk configuration: Users

- Ensure user is created

```
$ ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17
```

```
PLAY [Install Checkmk server] *****
```

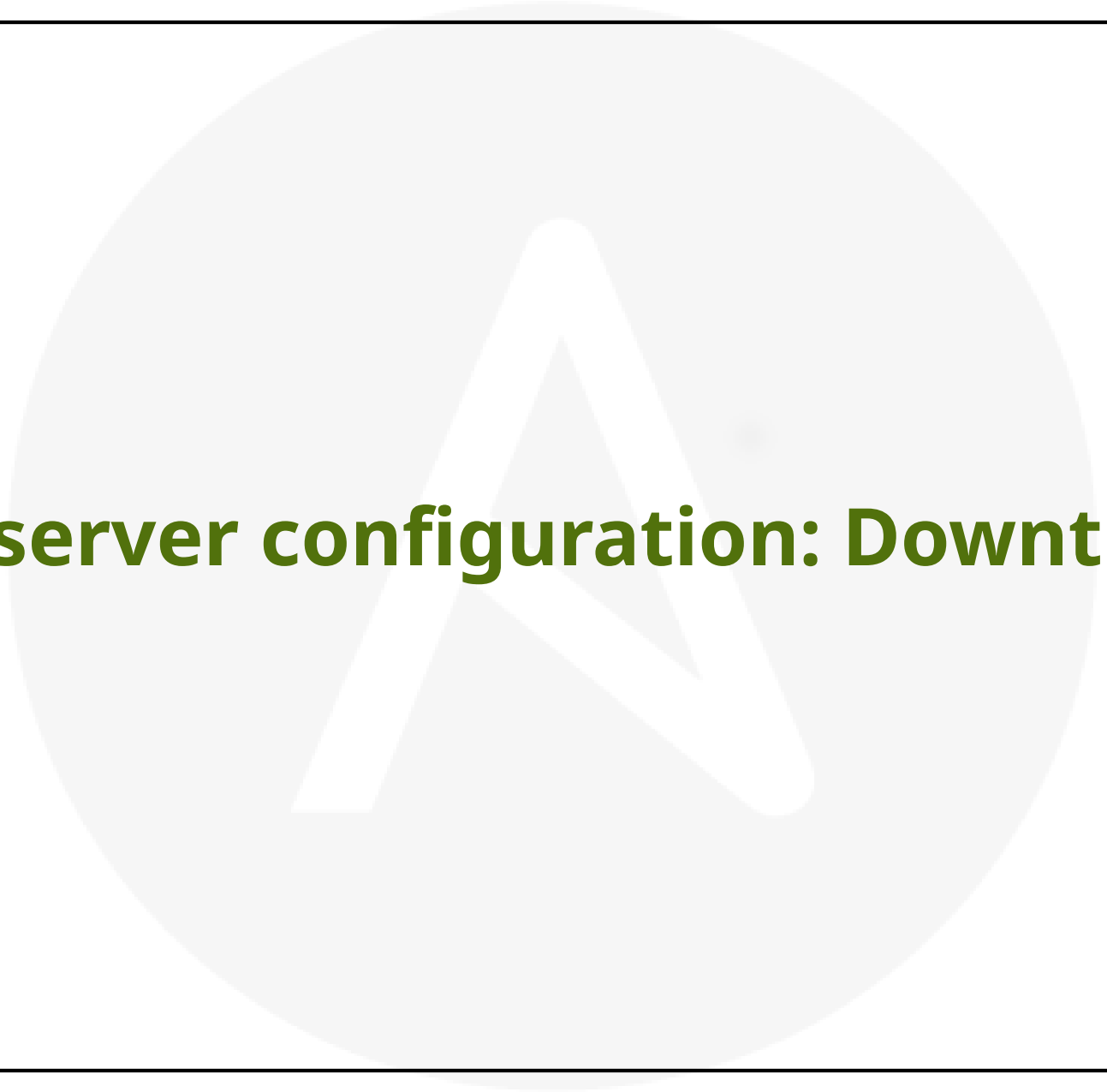
```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=19 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



# Checkmk server configuration: Downtimes

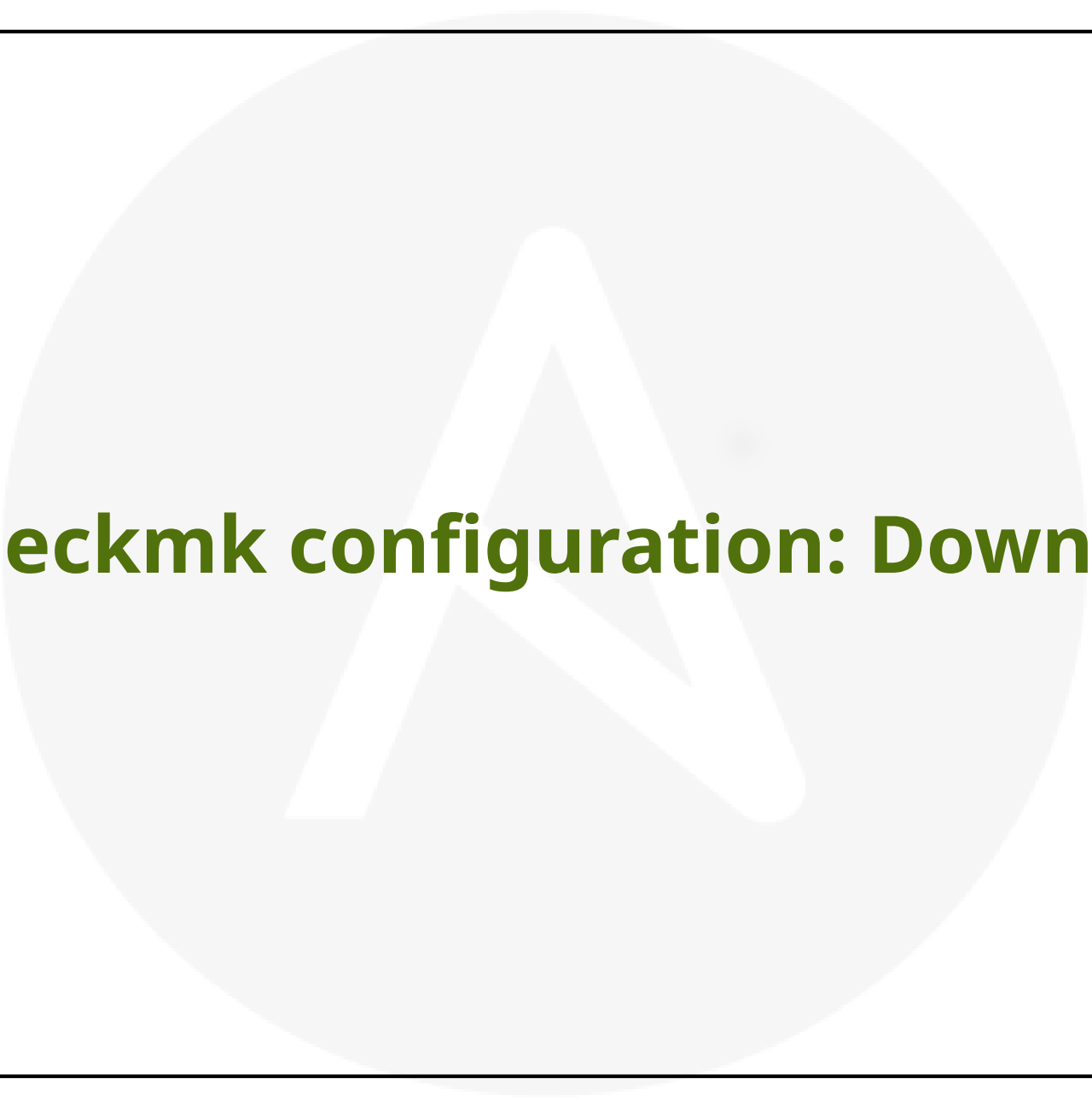
# Checkmk configuration: Downtimes

- Create downtime for **15 minutes**

```
<EDITOR> cmk_downtime.yml
```

# Checkmk configuration: Downtimes

```
---  
  
- name: Configure Checkmk downtime  
  hosts: cmk-agent  
  connection: local  
  become: true  
  
  tasks:  
    - name: Create downtime  
      checkmk.general.downtime:  
        host_name: "{{ inventory_hostname }}"  
        end_after:  
          minutes: 15  
        state: present  
        server_url: "https://localhost"  
        site: "master"  
        automation_user: "cmkadmin"  
        automation_secret: "{{ checkmk_agent_pass }}"  
        validate_certs: false
```



# LAB 11: Checkmk configuration: Downtimes

# LAB 11: Checkmk configuration: Downtimes

- Ensure downtime is created

```
$ ansible-playbook -i hosts cmk_downtime.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17
```

```
PLAY [Configure Checkmk downtime] *****
```

```
TASK [Gathering Facts] *****
```

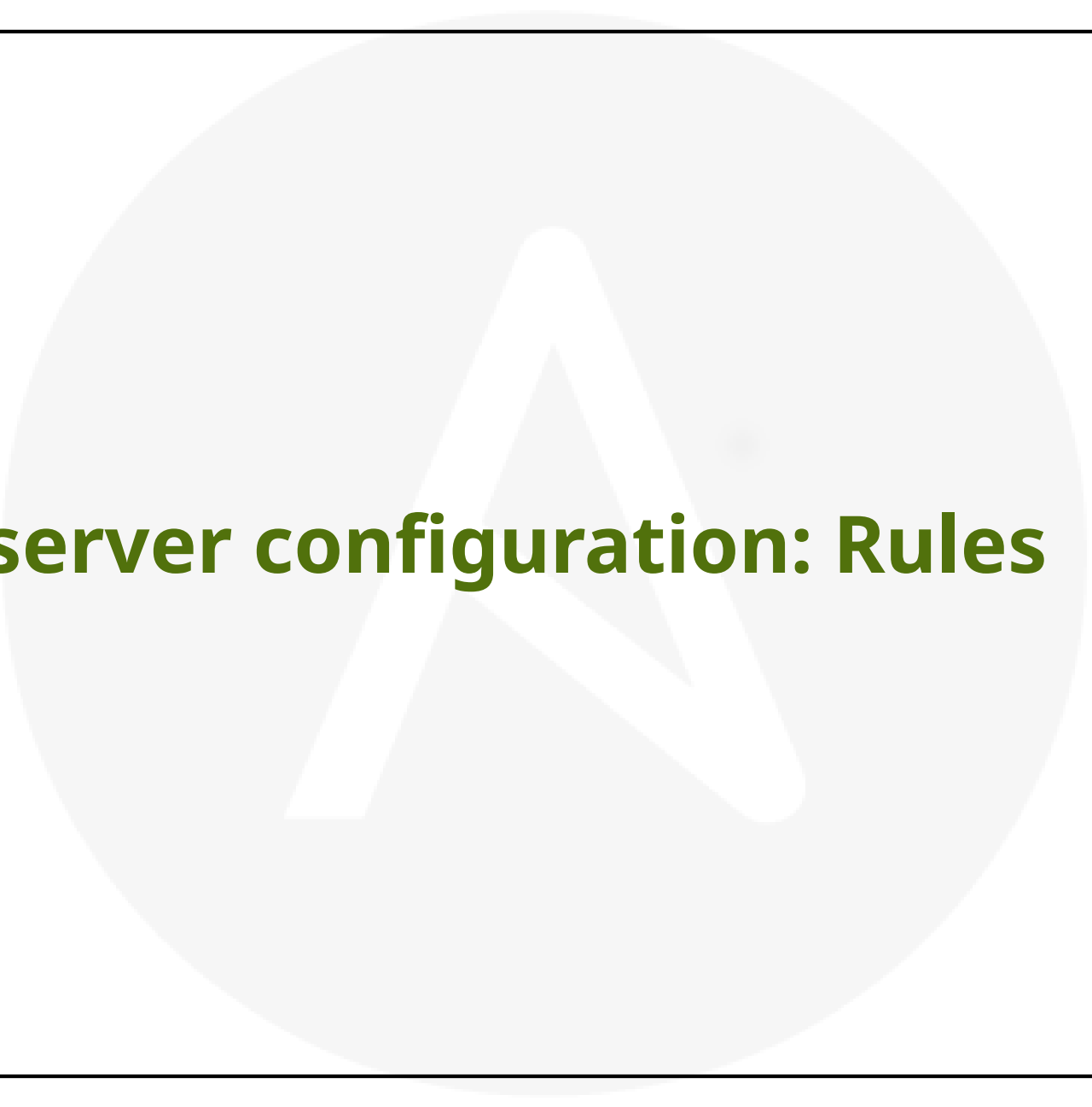
```
ok: [cmk-agent]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-agent : ok=3 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
```





# Checkmk server configuration: Rules

# Checkmk configuration: Rules

- Create a rule which set's **Check\_MK Agent** Check to **warning** if version is **different** from master, and **critical** if **TLS** is **not activated**

```
<EDITOR> host_vars/cmk-server.yml
```

# Checkmk configuration: Rules

```
...
checkmk_rules:
- ruleset: "checkgroup_parameters:agent_update"
  rule:
    properties:
      description: "Checkmk Agent version"
    location:
      folder: "/linux"
    value_raw: "{ 'agent_version': ('site', {}), 'agent_version_mismatch': 1, 'legacy_pull_mode': 2}"
```

## NOTE

Create rule in WebUI first, and access API to get syntax for rule values!

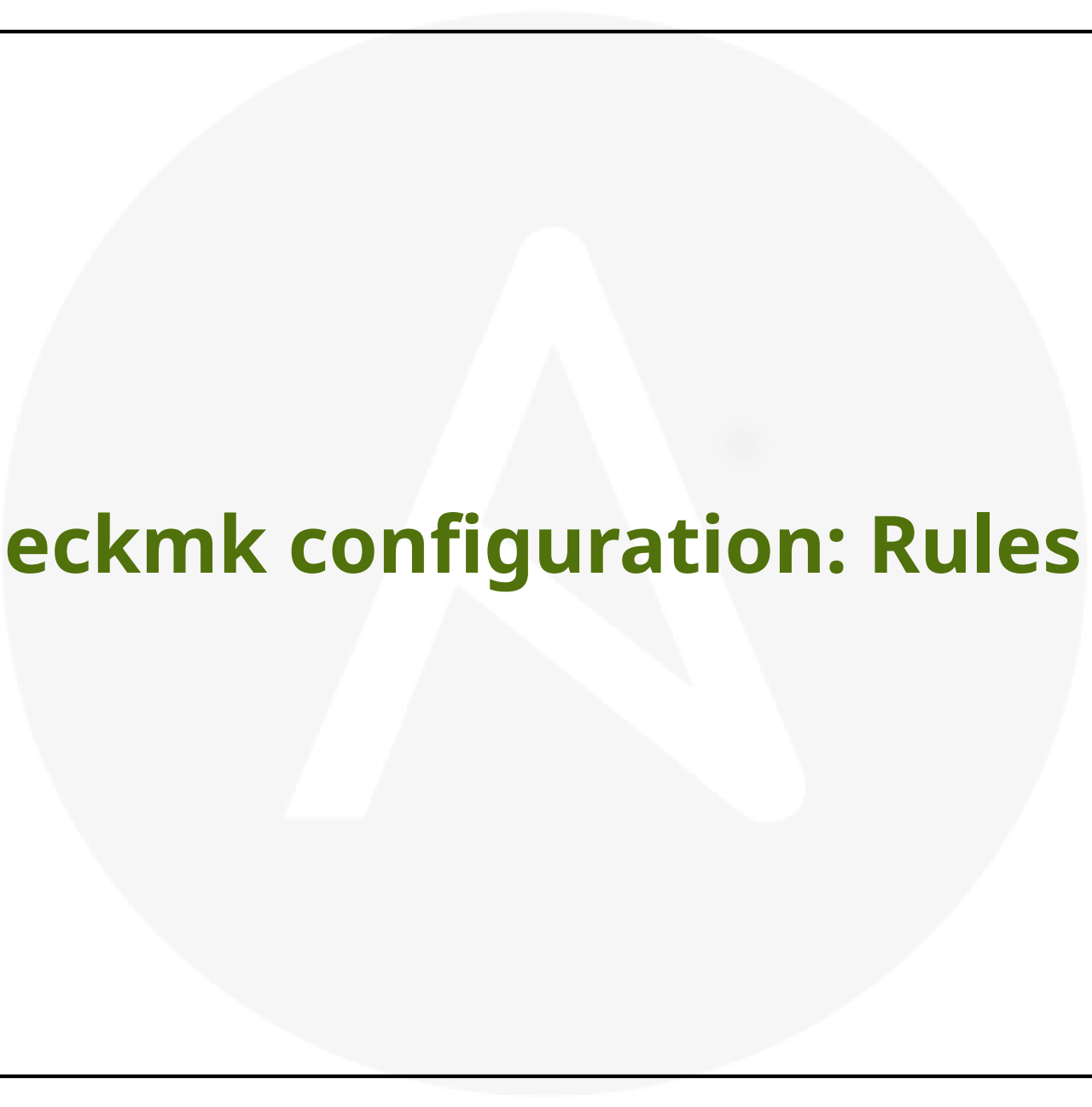
# Checkmk configuration: Rules

- Create a rule which set's **Check\_MK Agent** Check to **warning** if version is **different** from master, and **critical** if **TLS** is **not activated**
- Add task after existing tasks

```
$ <EDITOR> cmk_server.yml
```

# Checkmk configuration: Rules

```
...  
- name: Create rule  
  checkmk.general.rule:  
    rule: "{{ item['rule'] }}"  
    ruleset: "{{ item['ruleset'] }}"  
    state: present  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"  
    loop: "{{ checkmk_rules }}"
```



# LAB 12: Checkmk configuration: Rules

# LAB 12: Checkmk configuration: Rules

- Ensure rule for agent monitoring is created

```
ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17
```

```
PLAY [Install Checkmk server] *****
```

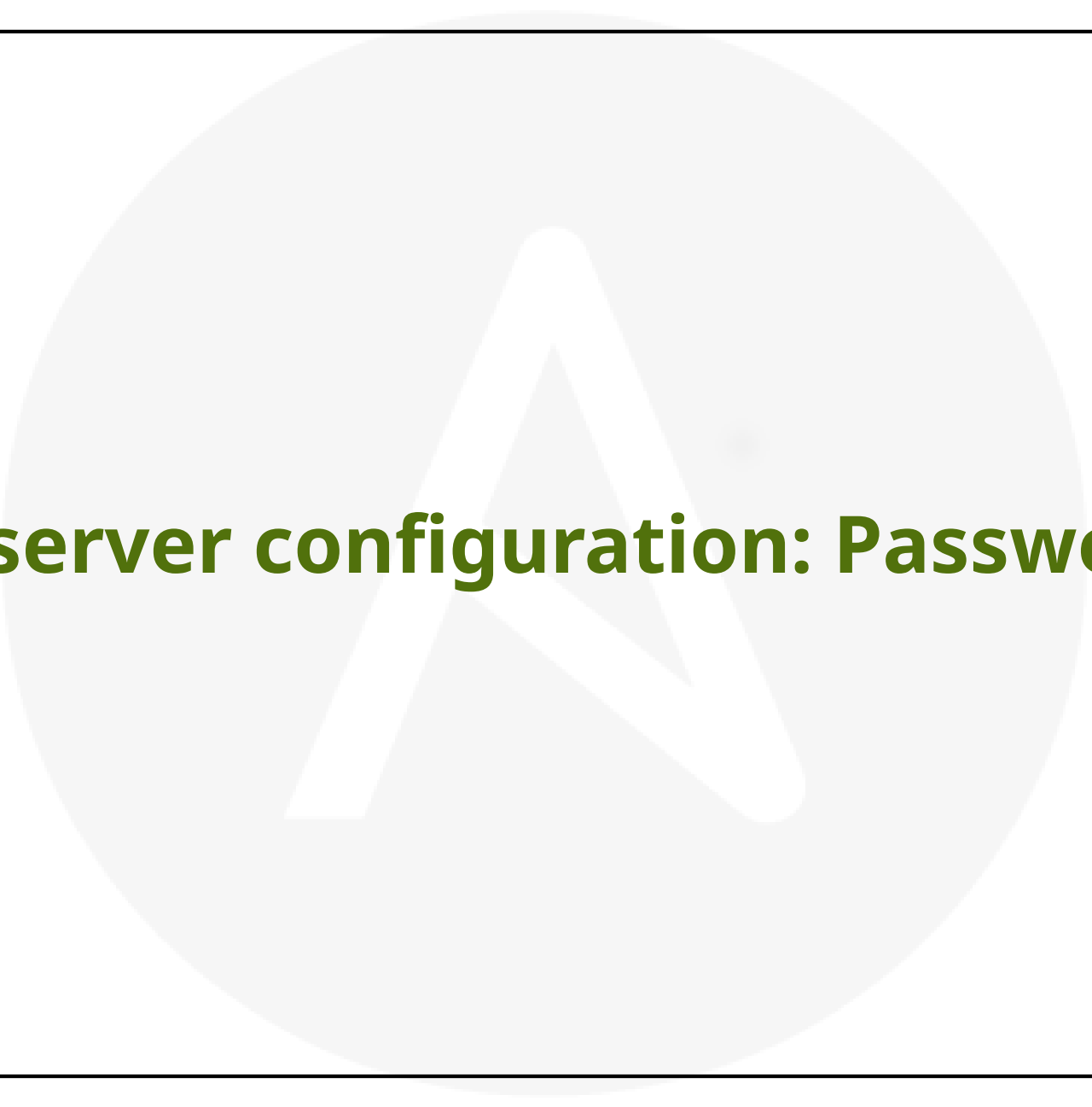
```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=20 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



# Checkmk server configuration: Passwords



# Checkmk configuration: Passwords

- Create a password for site **authenticationtest.com**

```
<EDITOR> host_vars/cmkn-server.yml
```

# Checkmk configuration: Passwords

```
...
```

```
checkmk_passwords:  
- name: authenticationtest_com  
  title: authenticationtest.com  
  password: pass
```

# Checkmk configuration: Passwords

- Create a password for site **authenticationtest.com**
- Add tasks **before Create rule** task!

```
<EDITOR> host_vars/cmkn-server.yml
```

# Checkmk configuration: Passwords

```
...  
  
- name: Create password  
  checkmk.general.password:  
    name: "{{ item['name'] }}"  
    title: "{{ item['title'] | default(omit) }}"  
    password: "{{ item['password'] }}"  
    state: present  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"  
    loop: "{{ checkmk_passwords }}"
```

# Checkmk configuration: Passwords

- Create a password for site **authenticationtest.com**
- Add rule to existing rules

```
<EDITOR> host_vars/cmkn-server.yml
```

# Checkmk configuration: Passwords

```
...
checkmk_rules:
...
- ruleset: "active_checks:httpv2"
  rule:
    properties:
      description: "authenticationtest.com"
    location:
      folder: "/"
    value_raw: >
      "{ 'endpoints': [{ 'service_name': { 'prefix': 'auto', 'name': 'HTTPAuth'}, 'url': 'https://authenticationtest.com/HTTPAuth/' }],
        'standard_settings': { 'connection': { 'method': ('get', None), 'auth': ('user_auth', { 'user': 'user', 'password':
          ('cmk_postprocessed', 'stored_password', ('authenticationtest_com', ''))) } } } } }"
    conditions:
      host_name:
        match_on:
          - cmk-agent
        operator: one_of
```

## NOTE

Create rule in WebUI first, and access API to get syntax for rule values!



# LAB 13: Checkmk configuration: Passwords

# LAB 13: Checkmk configuration: Passwords

- Ensure rule and password for web monitoring is created

```
$ ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

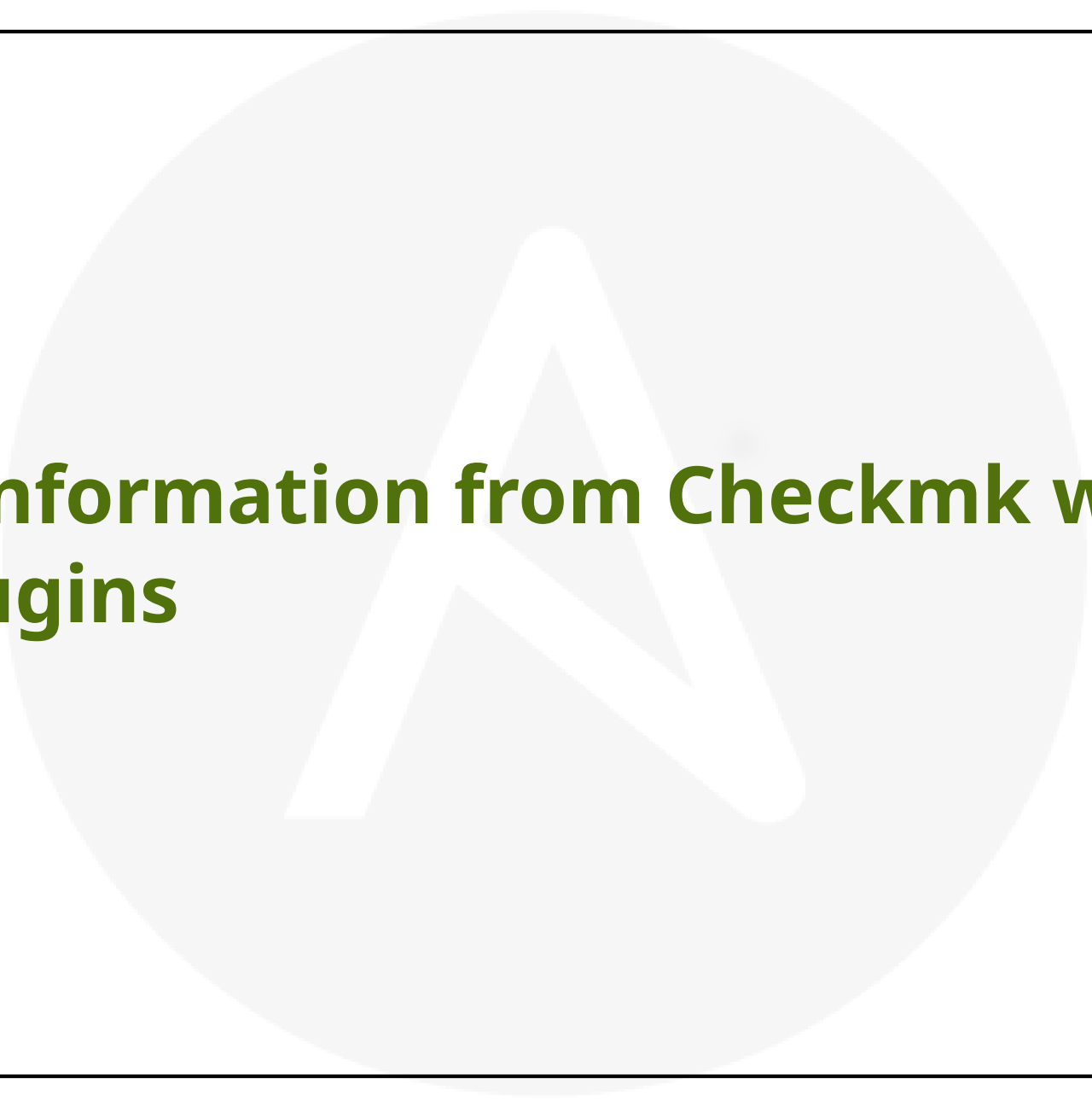
```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=21 changed=4 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```







# Fetching information from Checkmk with lookup plugins

# Checkmk lookup

- Print **server version** of Checkmk server
- Add task after existing tasks

```
<EDITOR> cmk_server.yml
```

# Checkmk lookups

...

- name: Print Checkmk server version

ansible.builtin.debug:

```
msg: "Checkmk server version is {{ lookup('checkmk.general.version',  
server_url='https://localhost',  
site='master',  
automation_user='cmkadmin',  
automation_secret=checkmk_admin_pw,  
validate_certs=false)}}
```

```
}}"
```



# LAB 14: Checkmk lookups

# LAB 14: Checkmk lookups

- Print Checkmk server version

```
$ ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
TASK [Print Checkmk server version] *****
```

```
ok: [cmk-server] => {
```

```
  "msg": "Checkmk server version is 2.5.0.cre"
```

```
}
```

```
PLAY RECAP *****
```

```
cmk-server : ok=22 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



# Using Checkmk as Ansible inventory

# Checkmk as Ansible inventory

- Checkmk **should** be aware of all hosts
- Can be a good source for Ansible
- Most likely requires configuration of tags



# Checkmk as Ansible inventory

- Create tags for **test** and **production** hosts

```
<EDITOR> host_vars/cmkn-server.yml
```

# Checkmk as Ansible inventory

```
...
```

```
checkmk_tag_groups:  
  - name: lifecycle_environment  
    title: Lifecycle Environment  
    tags:  
      - id: production  
        title: Production  
      - id: testing  
        title: Testing
```

# Checkmk as Ansible inventory

- Create tags for **test** and **production** hosts
- Add task after existing tasks

```
<EDITOR> host_vars/cmkn-server.yml
```

# Checkmk as Ansible inventory

```
...  
- name: Create tag group  
  checkmk.general.tag_group:  
    name: "{{ item['name'] }}"  
    title: "{{ item['title'] }}"  
    tags: "{{ item['tags'] }}"  
    state: present  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"  
    loop: "{{ checkmk_tag_groups }}"
```

# Checkmk as Ansible inventory

- Create inventory config for Checkmk

```
<EDITOR> checkmk.yml
```

## NOTE

Inventory plugin must end with checkmk.yml or checkmk.yaml!

# Checkmk as Ansible inventory

```
---
```

```
plugin: checkmk.general.checkmk
server_url: https://localhost
site: master
automation_user: cmkadmin
automation_secret: "AnsibleW0rkshop2026!"
validate_certs: false
groupsources:
  - hosttags
  - sites
```



# LAB 15: Checkmk as Ansible inventory

# LAB 15: Checkmk as Ansible inventory

- Ensure tag groups are created

```
ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.14.17
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=23 changed=4 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



# LAB 15: Checkmk as Ansible inventory

- Print all hosts and groups

```
ansible-inventory -i checkmk.yml --graph
```

```
@all:
  |--@ungrouped:
  |--@tag_criticality_prod:
  |   |--cmk-agent
  ...
  |--@tag_lifecycle_environment_production:
  |   |--cmk-agent
  |--@tag_lifecycle_environment_testing:
  ...
  |--@tag_lifecycle_environment_production:
  |   |--cmk-agent
  |--@tag_lifecycle_environment_testing:
```

# LAB 15: Checkmk as Ansible inventory

- Print host details

```
ansible-inventory -i checkmk.yml --host cmk-agent
```

```
{
  "checkmk_agent_edition": "cre",
  "checkmk_agent_pass": "AnsibleW0rkshop2026!",
  "checkmk_agent_server": "localhost",
  "checkmk_agent_server_validate_certs": "false",
  "checkmk_agent_site": "master",
  "checkmk_agent_user": "cmkadmin",
  "checkmk_agent_version": "2.5.0",
  "checkmk_host_settings": {
    "attributes": {
      "ipaddress": "127.0.0.1"
    },
    "folder": "/linux/checkmk"
  },
  "folder": "/linux/checkmk",
  "ipaddress": "127.0.0.1"
}
```



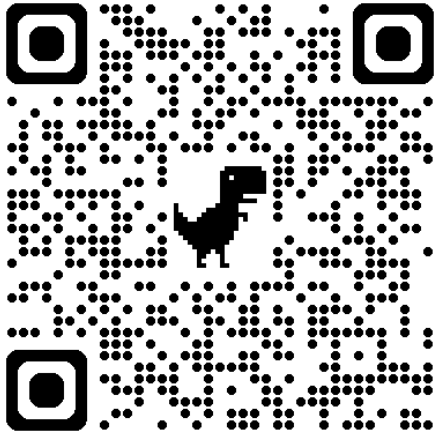
# Demo: Self healing with Checkmk, Ansible Automation Platform and Rule books

# Feedback



# Feedback

- <https://forms.gle/ki6KjQF9KS6BavmU8>



# Thanks a lot for your participation

- DI (FH) René Koch
- Freelancer
- The Checkmk Conference #11, 22.05.2025